

T4 Redes neuronales 2

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1 Introducción

Objetivo: introducir el algoritmo backprop para entrenar perceptrones multicapa con ejemplos de uso, así como algunas arquitecturas destacadas del aprendizaje profundo para clasificación de imágenes

Backprop: primero veremos el algoritmo backprop genérico para hallar el gradiente de un modelo

Ejemplo con una red para regresión: cálculo del gradiente y actualización de parámetros

- A partir de un dato
- Generalización de dato a batch

XOR: cálculo del gradiente y actualización de parámetros

- Ejemplo: red con una capa oculta de dos ReLUs y salida softmax
- Ejercicio: análogo al ejemplo, pero con salida sigmoide en lugar de softmax

2 Algoritmo backprop

Algoritmo backprop (backpropagation):

- Calcula el gradiente de la pérdida con respecto a los parámetros de cada capa
- El gradiente calculado se pasa a un algoritmo de optimización
- Origen: 1982; Paul Werbos lo aplica a MLPs de la manera que ha pasado a ser estándar
- Popularización: Rumelhart, Hinton y Williams lo popularizan en un artículo de Nature de 1986

Entrada: MLP de $L - 1$ capas, cada una con su $\theta^{(\ell)}$ ($\emptyset$ si no tiene), y pérdida respecto a $(\mathbf{x}, \mathbf{y})$ en "capa" L

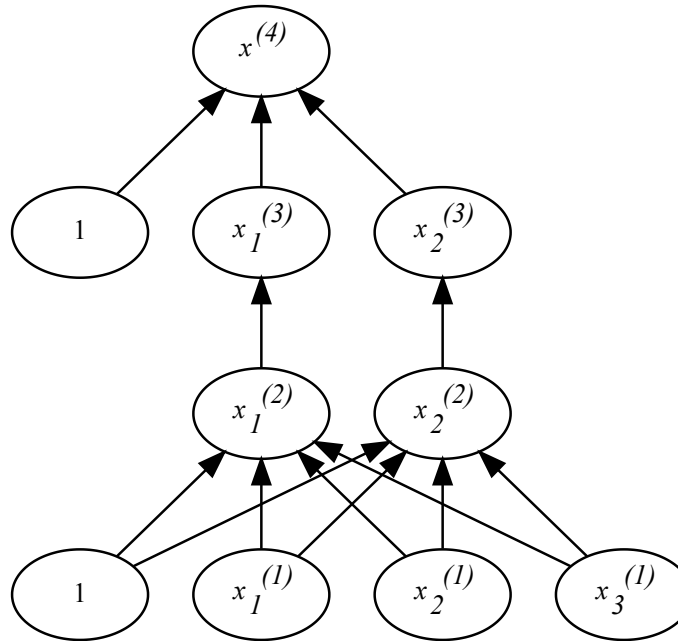
$$\mathbf{x}^{(1)} = \mathbf{x} \rightarrow \mathbf{x}^{(2)} = \mathbf{f}^{(1)}(\mathbf{x}^{(1)}, \theta^{(1)}) \rightarrow \dots \rightarrow \mathbf{x}^{(L)} = \mathbf{f}^{(L-1)}(\mathbf{x}^{(L-1)}, \theta^{(L-1)}) \rightarrow \mathbf{x}^{(L+1)} = \mathbf{f}^{(L)}(\mathbf{x}^{(L)}, \mathbf{y}) = \mathcal{L}$$

Método: bucle que ejecuta dos pasos en cada iteración

- **Forward:** recorre el modelo hacia adelante para hallar la salida de cada capa a partir de una entrada $\mathbf{x}$
 - $\mathbf{x}^{(1)} = \mathbf{x}$
 - para $\ell = 1 : L :$ $\mathbf{x}^{(\ell+1)} = \mathbf{f}^{(\ell)}(\mathbf{x}^{(\ell)}, \theta^{(\ell)})$
- **Backward:** recorre el modelo hacia atrás para hallar el gradiente de la pérdida respecto a los parámetros
 - $u^{(L+1)} = 1$
 - para $\ell = L : 1 :$ $\mathbf{g}^{(\ell)} = \mathbf{u}^{(\ell+1)t} \frac{\partial \mathbf{x}^{(\ell+1)}}{\partial \theta^{(\ell)}} \quad \mathbf{u}^{(\ell)t} = \mathbf{u}^{(\ell+1)t} \frac{\partial \mathbf{x}^{(\ell+1)}}{\partial \mathbf{x}^{(\ell)}}$

3 Ejemplo de regresión con un dato

```
In [ ]: import graphviz; graphviz.Source('''digraph {rankdir=BT; splines=false; node[shape=oval fontsize=11]
{rank=same x10[label=1] x11[label=<<i>x<sub>1</sub><sup>(1)</sup></i>>]
  x12[label=<<i>x<sub>2</sub><sup>(1)</sup></i>>] x13[label=<<i>x<sub>3</sub><sup>(1)</sup></i>>]}
{rank=same x21[label=<<i>x<sub>1</sub><sup>(2)</sup></i>>] x22[label=<<i>x<sub>2</sub><sup>(2)</sup></i>>]}
{rank=same x30[label=1] x31[label=<<i>x<sub>1</sub><sup>(3)</sup></i>>]
  x32[label=<<i>x<sub>2</sub><sup>(3)</sup></i>>] y[label=<<i>x<sup>(4)</sup></i>>]}
{x10 x11 x12 x13} -> {x21 x22} x21 -> x31 x22 -> x32 {x30 x31 x32} -> y edge[style=invis]
x10->x11->x12->x13 x21->x22 x30->x31->x32}''').render(filename='3_Ejemplo_un_dato', format='svg');
```



$$\mathbf{x}^{(1)} \in \mathbb{R}^3 \rightarrow \mathbf{x}^{(2)} = \mathbf{W}^{(1)}\mathbf{x}^{(1)} + \mathbf{b}^{(1)} \in \mathbb{R}^2 \rightarrow \mathbf{x}^{(3)} = \sigma(\mathbf{x}^{(2)}) \in \mathbb{R}^2 \rightarrow x^{(4)} = \mathbf{W}^{(3)}\mathbf{x}^{(3)} + b^{(3)} \in \mathbb{R}$$

Objetivo: dado $(\mathbf{x}, y) = (1, 1)$, minimizar la pérdida cuadrática mediante una iteración de GD con $\eta = 0.1$

Parámetros actuales: supongamos que $\theta = \mathbf{1}$, o, siguiendo el formato del modelo,

$$\mathbf{W}^{(1)} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad \mathbf{b}^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \mathbf{W}^{(3)} = (1, 1) \quad b^{(3)} = 1$$

Pérdida: para hallar el gradiente de la pérdida cuadrática, consideramos la pérdida como una "capa" adicional

$$\mathbf{x}^{(1)} = \mathbf{x} \rightarrow \mathbf{x}^{(2)} \rightarrow \mathbf{x}^{(3)} \rightarrow x^{(4)} = \hat{y} \rightarrow x^{(5)} = \mathcal{L} = (y - \hat{y})^2$$

Forward: recorreremos el modelo hacia adelante para hallar la salida de cada capa a partir de la entrada

$$\mathbf{x}^{(1)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\mathbf{x}^{(2)} = \mathbf{W}^{(1)}\mathbf{x}^{(1)} + \mathbf{b}^{(1)} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

$$\mathbf{x}^{(3)} = \sigma(\mathbf{x}^{(2)}) = \sigma\left(\begin{pmatrix} 4 \\ 4 \end{pmatrix}\right) = \begin{pmatrix} \sigma(4) \\ \sigma(4) \end{pmatrix} = \begin{pmatrix} 0.982 \\ 0.982 \end{pmatrix}$$

$$\hat{y} = x^{(4)} = \mathbf{W}^{(3)}\mathbf{x}^{(3)} + b^{(3)} = (1, 1) \begin{pmatrix} 0.982 \\ 0.982 \end{pmatrix} + 1 = 2.9640$$

$$\mathcal{L} = x^{(5)} = (y - \hat{y})^2 = (1 - 2.9640)^2 = 3.8574$$

Backward: recorremos el modelo hacia atrás para hallar el gradiente de la pérdida respecto a los parámetros

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = 2(\hat{y} - y) = 3.9280$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial \mathbf{W}^{(3)}} = \mathbf{x}^{(3)} \frac{\partial \mathcal{L}}{\partial \hat{y}} = \begin{pmatrix} 3.8574 \\ 3.8574 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot 1 = 3.9280$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial \mathbf{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \mathbf{W}^{(3)} = (3.9280, 3.9280)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} \frac{\partial \mathbf{x}^{(3)}}{\partial \mathbf{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} \odot \boldsymbol{\sigma}'(\mathbf{x}^{(2)t}) = (0.0694, 0.0694)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \frac{\partial \mathbf{x}^{(2)}}{\partial \mathbf{W}^{(1)}} = \mathbf{x}^{(1)} \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} = \begin{pmatrix} 0.0694 & 0.0694 \\ 0.0694 & 0.0694 \\ 0.0694 & 0.0694 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \frac{\partial \mathbf{x}^{(2)}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \odot \mathbf{1}^t = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} = (0.0694, 0.0694)$$

Actualización de parámetros: lo hacemos capa a capa, respetando el formato del modelo

$$\begin{aligned}\mathbf{W}^{(3)} &= \mathbf{W}^{(3)} - \eta \left(\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} \right)^t \\ &= (1, 1) - 0.1 (3.8574, 3.8574) = (0.6143, 0.6143)\end{aligned}$$

$$\begin{aligned}b^{(3)} &= b^{(3)} - \eta \left(\frac{\partial \mathcal{L}}{\partial b^{(3)}} \right)^t \\ &= 1 - 0.1 \cdot 3.9280 = 0.6072\end{aligned}$$

$$\begin{aligned}\mathbf{W}^{(1)} &= \mathbf{W}^{(1)} - \eta \left(\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} \right)^t \\ &= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} - 0.1 \begin{pmatrix} 0.0694 & 0.0694 & 0.0694 \\ 0.0694 & 0.0694 & 0.0694 \end{pmatrix} = \begin{pmatrix} 0.9931 & 0.9931 & 0.9931 \\ 0.9931 & 0.9931 & 0.9931 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\mathbf{b}^{(1)} &= \mathbf{b}^{(1)} - \eta \left(\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} \right)^t \\ &= \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 0.1 \begin{pmatrix} 0.0694 \\ 0.0694 \end{pmatrix} = \begin{pmatrix} 0.9931 \\ 0.9931 \end{pmatrix}\end{aligned}$$

Solución a máquina: `keras` mantiene transpuestas de matrices de pesos y vectores de sesgos

```
In [ ]: import numpy as np; np.set_printoptions(precision=4); import keras
x1 = keras.Input(shape=(3,)); C1 = keras.initializers.Constant(1.)
x3 = keras.layers.Dense(2, activation='sigmoid', kernel_initializer=C1, bias_initializer=C1)(x1)
hy = keras.layers.Dense(1, activation=None, kernel_initializer=C1, bias_initializer=C1)(x3)
M = keras.Model(inputs=x1, outputs=hy); M.summary()
```

Model: "functional"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 3)	0
dense (Dense)	(None, 2)	8
dense_1 (Dense)	(None, 1)	3

Total params: 11 (44.00 B)

Trainable params: 11 (44.00 B)

Non-trainable params: 0 (0.00 B)

```
In [ ]: X = np.array([[1, 1, 1]]); y = np.array([[1]])
M_x3 = keras.Model(inputs=x1, outputs=M.layers[1].output); print('x3 =', M_x3(X))
print('hy =', M(X))
M.compile(loss='mean_squared_error', optimizer=keras.optimizers.SGD(learning_rate=0.1))
M.fit(X, y, epochs=1, verbose=1); print(M.layers[1].get_weights(), "\n", M.layers[2].get_weights())

x3 = tf.Tensor([[0.982 0.982]], shape=(1, 2), dtype=float32)
hy = tf.Tensor([[2.964]], shape=(1, 1), dtype=float32)
1/1 _____ 0s 292ms/step - loss: 3.8574
[array([[0.9931, 0.9931],
        [0.9931, 0.9931],
        [0.9931, 0.9931]], dtype=float32), array([0.9931, 0.9931], dtype=float32)]
[array([[0.6143],
        [0.6143]], dtype=float32), array([0.6072], dtype=float32)]
```

4 Ejemplo de regresión con un batch

Objetivo: entrenar el modelo anterior con $N = 2$ datos, $\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^t \\ \mathbf{x}_2^t \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}$ y $\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

Forward: la generalización de dato a batch es sencilla

$$\mathbf{X}^{(1)} = \mathbf{X} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}$$

$$\mathbf{X}^{(2)} = \mathbf{X}^{(1)} \mathbf{W}^{(1)t} + \mathbf{1} \mathbf{b}^{(1)t} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1, 1) = \begin{pmatrix} 4 & 4 \\ 7 & 7 \end{pmatrix}$$

$$\mathbf{X}^{(3)} = \sigma(\mathbf{X}^{(2)}) = \sigma \left(\begin{pmatrix} 4 & 4 \\ 7 & 7 \end{pmatrix} \right) = \begin{pmatrix} \sigma(4) & \sigma(4) \\ \sigma(7) & \sigma(7) \end{pmatrix} = \begin{pmatrix} 0.9820 & 0.9820 \\ 0.9991 & 0.9991 \end{pmatrix}$$

$$\hat{\mathbf{y}} = \mathbf{X}^{(4)} = \mathbf{X}^{(3)} \mathbf{W}^{(3)t} + \mathbf{1} \mathbf{b}^{(3)t} = \begin{pmatrix} 0.9820 & 0.9820 \\ 0.9991 & 0.9991 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} 1 = \begin{pmatrix} 2.9640 \\ 2.9982 \end{pmatrix}$$

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2.9640 \\ 2.9982 \end{pmatrix} = \begin{pmatrix} -1.9640 \\ -0.9982 \end{pmatrix}$$

$$\mathcal{L} = x^{(5)} = \frac{1}{N} \sum_n (y_n - \hat{y}_n)^2 = \frac{1}{N} (\mathbf{y} - \hat{\mathbf{y}})^t (\mathbf{y} - \hat{\mathbf{y}}) = 2.4268$$

Matrices de pesos y vectores de sesgos transpuestos: mayor eficiencia en forward (e inferencia)

Backward: generalización de dato a batch

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = 2(\hat{y} - y)$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} = \frac{2}{N}(\hat{\mathbf{y}} - \mathbf{y})^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial \mathbf{W}^{(3)}} = \mathbf{x}^{(3)} \frac{\partial \mathcal{L}}{\partial \hat{y}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t} \left(\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial b^{(3)}}$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \frac{\partial \hat{\mathbf{y}}}{\partial b^{(3)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial \mathbf{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \mathbf{W}^{(3)}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t} \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} \frac{\partial \mathbf{x}^{(3)}}{\partial \mathbf{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} \odot \boldsymbol{\sigma}'(\mathbf{x}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \frac{\partial \mathbf{X}^{(3)}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \boldsymbol{\sigma}'(\mathbf{X}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \frac{\partial \mathbf{x}^{(2)}}{\partial \mathbf{W}^{(1)}} = \mathbf{x}^{(1)} \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \frac{\partial \mathbf{x}^{(2)}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \odot \mathbf{1}^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \mathbf{b}^{(1)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

Backward: cálculos

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} = \frac{2}{N} (\hat{\mathbf{y}} - \mathbf{y})^t = (1.9640, 0.9982)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t} \left(\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \right)^t = \begin{pmatrix} 0.9820 & 0.9991 \\ 0.9820 & 0.9991 \end{pmatrix} \begin{pmatrix} 1.9640 \\ 0.9982 \end{pmatrix} = \begin{pmatrix} 2.9259 \\ 2.9259 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \right)^t = (1, 1) \begin{pmatrix} 1.9640 \\ 0.9982 \end{pmatrix} = 2.9622$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t} \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1.9640, 0.9982) = \begin{pmatrix} 1.9640 & 0.9982 \\ 1.9640 & 0.9982 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \boldsymbol{\sigma}'(\mathbf{X}^{(2)t}) = \begin{pmatrix} 1.9640 & 0.9982 \\ 1.9640 & 0.9982 \end{pmatrix} \odot \begin{pmatrix} \sigma'(4) & \sigma'(7) \\ \sigma'(4) & \sigma'(7) \end{pmatrix} = \begin{pmatrix} 0.0347 & 0.0009 \\ 0.0347 & 0.0009 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t = \begin{pmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0.0347 & 0.0347 \\ 0.0009 & 0.0009 \end{pmatrix} = \begin{pmatrix} 0.0365 & 0.0365 \\ 0.0365 & 0.0365 \\ 0.0365 & 0.0365 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial b^{(1)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t = (1, 1) \begin{pmatrix} 0.0347 & 0.0347 \\ 0.0009 & 0.0009 \end{pmatrix} = (0.0356, 0.0356)$$

Actualización de parámetros: capa a capa, respetando el formato (transpuesto) del modelo

$$\mathbf{W}^{(3)t} = \mathbf{W}^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 0.1 \begin{pmatrix} 2.9259 \\ 2.9259 \end{pmatrix} = \begin{pmatrix} 0.7074 \\ 0.7074 \end{pmatrix}$$

$$b^{(3)t} = b^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial b^{(3)}} = 1 - 0.1 \cdot 2.9622 = 0.7038$$

$$\mathbf{W}^{(1)t} = \mathbf{W}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} - 0.1 \begin{pmatrix} 0.0365 & 0.0365 \\ 0.0365 & 0.0365 \\ 0.0365 & 0.0365 \end{pmatrix} = \begin{pmatrix} 0.9964 & 0.9964 \\ 0.9964 & 0.9964 \\ 0.9964 & 0.9964 \end{pmatrix}$$

$$\mathbf{b}^{(1)t} = \mathbf{b}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = (1, 1) - 0.1(0.0356, 0.0356) = (0.9964, 0.9964)$$

Solución a máquina:

```
In [ ]: import numpy as np; np.set_printoptions(precision=4); import keras
X1 = keras.Input(shape=(3,)); C1 = keras.initializers.Constant(1.)
X3 = keras.layers.Dense(2, activation='sigmoid', kernel_initializer=C1, bias_initializer=C1)(X1)
hy = keras.layers.Dense(1, activation=None, kernel_initializer=C1, bias_initializer=C1)(X3)
M = keras.Model(inputs=X1, outputs=hy);

In [ ]: X = np.array([[1, 1, 1], [2, 2, 2]]); y = np.array([[1], [2]])
M_X3 = keras.Model(inputs=X1, outputs=M.layers[1].output); print('X3 =', M_X3(X))
print('hy =', M(X))
M.compile(loss='mean_squared_error', optimizer=keras.optimizers.SGD(learning_rate=0.1))
M.fit(X, y, epochs=1, verbose=1); print(M.layers[1].get_weights(), "\n", M.layers[2].get_weights())

X3 = tf.Tensor(
[[0.982  0.982 ]
 [0.9991 0.9991]], shape=(2, 2), dtype=float32)
hy = tf.Tensor(
[[2.964 ]
 [2.9982]], shape=(2, 1), dtype=float32)
1/1 _____ 0s 190ms/step - loss: 2.4269
[array([[0.9963, 0.9963],
        [0.9963, 0.9963],
        [0.9963, 0.9963]], dtype=float32), array([0.9964, 0.9964], dtype=float32)]
[array([[0.7074],
        [0.7074]], dtype=float32), array([0.7038], dtype=float32)]
```

5 Ejemplo para XOR con salida softmax

Modelo: capa oculta con dos ReLUs y capa de salida con softmax para dos clases

$$\mathbf{x}^{(1)} \in \{0, 1\}^2$$

$$\mathbf{x}^{(2)} = \mathbf{W}^{(1)}\mathbf{x}^{(1)} + \mathbf{b}^{(1)} \in \mathbb{R}^2$$

$$\mathbf{x}^{(3)} = \text{ReLU}(\mathbf{x}^{(2)}) \in \mathbb{R}^2$$

$$\mathbf{x}^{(4)} = \mathbf{W}^{(3)}\mathbf{x}^{(3)} + \mathbf{b}^{(3)} \in \mathbb{R}^2$$

$$\mathbf{x}^{(5)} = \hat{\mathbf{y}} = \mathcal{S}(\mathbf{x}^{(4)}) \in [0, 1]^2$$

$$\mathbf{x}^{(6)} = \mathcal{L} = \mathbb{H}(\mathbf{y}, \hat{\mathbf{y}}) = -\mathbf{y}^t \log(\hat{\mathbf{y}})$$

Parámetros actuales:

$$\mathbf{W}^{(1)} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \mathbf{b}^{(1)} = \begin{pmatrix} -1 \\ 0.5 \end{pmatrix} \quad \mathbf{W}^{(3)} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad \mathbf{b}^{(3)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Objetivo: backprop con $\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^t \\ \mathbf{x}_2^t \\ \mathbf{x}_3^t \\ \mathbf{x}_4^t \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}$ y $\mathbf{Y} = \begin{pmatrix} \mathbf{y}_1^t \\ \mathbf{y}_2^t \\ \mathbf{y}_3^t \\ \mathbf{y}_4^t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$

Forward:

$$\mathbf{X}^{(2)} = \mathbf{X}^{(1)} \mathbf{W}^{(1)t} + \mathbf{1} \mathbf{b}^{(1)t} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} (-1, 0.5) = \begin{pmatrix} -1 & 0.5 \\ 0 & 1.5 \\ 0 & 1.5 \\ 1 & 2.5 \end{pmatrix}$$

$$\mathbf{X}^{(3)} = \text{ReLU}(\mathbf{X}^{(2)}) = \begin{pmatrix} 0 & 0.5 \\ 0 & 1.5 \\ 0 & 1.5 \\ 1 & 2.5 \end{pmatrix}$$

$$\mathbf{X}^{(4)} = \mathbf{X}^{(3)} \mathbf{W}^{(3)t} + \mathbf{1} \mathbf{b}^{(3)t} = \begin{pmatrix} 0 & 0.5 \\ 0 & 1.5 \\ 0 & 1.5 \\ 1 & 2.5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} (1, -1) = \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \\ -0.5 & 0.5 \\ -0.5 & 0.5 \end{pmatrix}$$

$$\hat{\mathbf{Y}} = \mathbf{X}^{(5)} = \mathcal{S}(\mathbf{X}^{(4)}) = \mathcal{S} \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \\ -0.5 & 0.5 \\ -0.5 & 0.5 \end{pmatrix} = \begin{pmatrix} 0.7311 & 0.2689 \\ 0.2689 & 0.7311 \\ 0.2689 & 0.7311 \\ 0.2689 & 0.7311 \end{pmatrix}$$

$$\mathcal{L} = x^{(6)} = -\frac{1}{N} \mathbf{1}^t (\mathbf{Y} \odot \log \hat{\mathbf{Y}}) \mathbf{1} = -\frac{1}{4} (1, 1, 1, 1) \begin{pmatrix} -0.3132 & 0 \\ 0 & -0.3132 \\ 0 & -0.3132 \\ -1.3134 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = 0.5633$$

Backward: para un dato y generalización a batch

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} = -\mathbf{y}^t \oslash \hat{\mathbf{y}}^t$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{Y}}} = -\frac{1}{N}(\mathbf{Y} \oslash \hat{\mathbf{Y}})^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(4)}} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{x}^{(4)}} = (\hat{\mathbf{y}} - \mathbf{y})^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{Y}}} \frac{\partial \hat{\mathbf{Y}}}{\partial \mathbf{X}^{(4)}} = \frac{1}{N}(\hat{\mathbf{Y}} - \mathbf{Y})^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(4)}} \frac{\partial \mathbf{x}^{(4)}}{\partial \mathbf{W}^{(3)}} = \mathbf{x}^{(3)} \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(4)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(4)}} \frac{\partial \mathbf{x}^{(4)}}{\partial \mathbf{b}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(4)}} \odot \mathbf{1}^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial \mathbf{b}^{(3)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(4)}} \frac{\partial \mathbf{x}^{(4)}}{\partial \mathbf{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(4)}} \mathbf{W}^{(3)}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t} \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} \frac{\partial \mathbf{x}^{(3)}}{\partial \mathbf{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} \odot \text{ReLU}'(\mathbf{x}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \frac{\partial \mathbf{X}^{(3)}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \text{ReLU}'(\mathbf{X}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \frac{\partial \mathbf{x}^{(2)}}{\partial \mathbf{W}^{(1)}} = \mathbf{x}^{(1)} \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \frac{\partial \mathbf{x}^{(2)}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \odot \mathbf{1}^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \mathbf{b}^{(1)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

Backward: cálculos

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \frac{1}{N} (\hat{\mathbf{Y}} - \mathbf{Y})^t = \begin{pmatrix} -0.0672 & 0.0672 & 0.0672 & -0.1828 \\ 0.0672 & -0.0672 & -0.0672 & 0.1828 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t = \begin{pmatrix} -0.1828 & 0.1828 \\ -0.2890 & 0.2890 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(3)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t = (-0.1156, 0.1156)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t} \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \begin{pmatrix} -0.1344 & 0.1344 & 0.1344 & -0.3656 \\ 0.1344 & -0.1344 & -0.1344 & 0.3656 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \text{ReLU}'(\mathbf{X}^{(2)t}) = \begin{pmatrix} 0 & 0 & 0 & -0.3656 \\ 0.1344 & -0.1344 & -0.1344 & 0.3656 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0.1344 \\ 0 & -0.1344 \\ 0 & -0.1344 \\ -0.3656 & 0.3656 \end{pmatrix} = \begin{pmatrix} -0.3656 & 0.2312 \\ -0.3656 & 0.2312 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t = (1, 1, 1, 1) \begin{pmatrix} 0 & 0.1344 \\ 0 & -0.1344 \\ 0 & -0.1344 \\ -0.3656 & 0.3656 \end{pmatrix} = (-0.3656, 0.2312)$$

Actualización de parámetros: capa a capa, respetando el formato (transpuesto) del modelo

$$\mathbf{W}^{(3)t} = \mathbf{W}^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} - 0.1 \begin{pmatrix} -0.1828 & 0.1828 \\ -0.2890 & 0.2890 \end{pmatrix} = \begin{pmatrix} 1.0183 & -1.0183 \\ -0.9711 & 0.9711 \end{pmatrix}$$

$$\mathbf{b}^{(3)t} = \mathbf{b}^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(3)}} = (1, -1) - 0.1(-0.1156, 0.1156) = (1.0116, -1.0116)$$

$$\mathbf{W}^{(1)t} = \mathbf{W}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - 0.1 \begin{pmatrix} -0.3656 & 0.2312 \\ -0.3656 & 0.2312 \end{pmatrix} = \begin{pmatrix} 1.0366 & 0.9769 \\ 1.0366 & 0.9769 \end{pmatrix}$$

$$\mathbf{b}^{(1)t} = \mathbf{b}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = (-1, 0.5) - 0.1(-0.3656, 0.2312) = (-0.9634, 0.4769)$$

Solución a máquina:

```
In [ ]: import numpy as np; np.set_printoptions(precision=4); import tensorflow as tf; import keras
X1 = keras.Input(shape=(2,))
W1 = tf.constant_initializer([[1, 1], [1, 1]]); b1 = tf.constant_initializer([[ -1], [.5]])
X3 = keras.layers.Dense(2, activation='relu', kernel_initializer=W1, bias_initializer=b1)(X1)
W3 = tf.constant_initializer([[1, -1], [-1, 1]]); b3 = tf.constant_initializer([[1], [-1]])
hY = keras.layers.Dense(2, activation='softmax', kernel_initializer=W3, bias_initializer=b3)(X3)
M = keras.Model(inputs=X1, outputs=hY);

In [ ]: X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]]); y = np.array([[1, 0], [0, 1], [0, 1], [1, 0]])
M_X3 = keras.Model(inputs=X1, outputs=M.layers[1].output); print('X3 =', M_X3(X))
print('hY =', M(X))
M.compile(loss='crossentropy', optimizer=keras.optimizers.SGD(learning_rate=0.1))
M.fit(X, y, epochs=1, verbose=1); print(M.layers[1].get_weights(), "\n", M.layers[2].get_weights())

X3 = tf.Tensor(
[[0.  0.5]
 [0.  1.5]
 [0.  1.5]
 [1.  2.5]], shape=(4, 2), dtype=float32)
hY = tf.Tensor(
[[0.7311 0.2689]
 [0.2689 0.7311]
 [0.2689 0.7311]
 [0.2689 0.7311]], shape=(4, 2), dtype=float32)
1/1 _____ 0s 294ms/step - loss: 0.5633
[array([[1.0366, 0.9769],
        [1.0366, 0.9769]], dtype=float32), array([-0.9634,  0.4769], dtype=float32)]
[array([[ 1.0183, -1.0183],
        [-0.9711,  0.9711]], dtype=float32), array([ 1.0116, -1.0116], dtype=float32)]
```

6 Ejercicio: XOR con salida sigmoide

Modelo: capa oculta con dos ReLUs y capa de salida con sigmoide para dos clases

$$\mathbf{x}^{(1)} \in \{0, 1\}^2$$

$$\mathbf{x}^{(2)} = \mathbf{W}^{(1)} \mathbf{x}^{(1)} + \mathbf{b}^{(1)} \in \mathbb{R}^2$$

$$\mathbf{x}^{(3)} = \text{ReLU}(\mathbf{x}^{(2)}) \in \mathbb{R}^2$$

$$\mathbf{x}^{(4)} = \mathbf{W}^{(3)} \mathbf{x}^{(3)} + \mathbf{b}^{(3)} \in \mathbb{R}$$

$$x^{(5)} = \hat{y} = \sigma(x^{(4)}) \in [0, 1]$$

$$x^{(6)} = \mathcal{L} = \mathbb{H}(y, \hat{y}) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

Parámetros actuales:

$$\mathbf{W}^{(1)} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \mathbf{b}^{(1)} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \quad \mathbf{W}^{(3)} = \begin{pmatrix} 1 & -1 \end{pmatrix} \quad b^{(3)} = 0$$

Objetivo: backprop con $\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^t \\ \mathbf{x}_2^t \\ \mathbf{x}_3^t \\ \mathbf{x}_4^t \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}$ y $\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$

Solución:

Forward:

$$\mathbf{X}^{(2)} = \mathbf{X}^{(1)} \mathbf{W}^{(1)t} + \mathbf{1} \mathbf{b}^{(1)t} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} (0, -1) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \\ 1 & 0 \\ 2 & 1 \end{pmatrix}$$

$$\mathbf{X}^{(3)} = \text{ReLU}(\mathbf{X}^{(2)}) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 2 & 1 \end{pmatrix}$$

$$\mathbf{X}^{(4)} = \mathbf{X}^{(3)} \mathbf{W}^{(3)t} + \mathbf{1} \mathbf{b}^{(3)t} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} 0 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\hat{\mathbf{y}} = \mathbf{X}^{(5)} = \boldsymbol{\sigma}(\mathbf{X}^{(4)}) = \begin{pmatrix} 0.5000 \\ 0.7310 \\ 0.7310 \\ 0.7310 \end{pmatrix}$$

$$\mathcal{L} = x^{(6)} = -\frac{1}{N}(\mathbf{y}^t \log(\hat{\mathbf{y}}) + (\mathbf{1} - \mathbf{y})^t \log(\mathbf{1} - \hat{\mathbf{y}})) = -\frac{1}{4}(2 \log 0.7310 + \log 0.5 + \log 0.2690) = 0.6582$$

Backward: para un dato y generalización a batch

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})}$$

$$\frac{\partial \mathcal{L}}{\partial x^{(4)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial x^{(4)}} = \hat{y} - y$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial x^{(4)}} \frac{\partial x^{(4)}}{\partial \mathbf{W}^{(3)}} = \mathbf{x}^{(3)} \frac{\partial \mathcal{L}}{\partial x^{(4)}}$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial x^{(4)}} \frac{\partial x^{(4)}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial x^{(4)}} \odot \mathbf{1}^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial x^{(4)}} \frac{\partial x^{(4)}}{\partial \mathbf{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial x^{(4)}} \mathbf{W}^{(3)}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} \frac{\partial \mathbf{x}^{(3)}}{\partial \mathbf{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(3)}} \odot \text{ReLU}'(\mathbf{x}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \frac{\partial \mathbf{x}^{(2)}}{\partial \mathbf{W}^{(1)}} = \mathbf{x}^{(1)} \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \frac{\partial \mathbf{x}^{(2)}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^{(2)}} \odot \mathbf{1}^t$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} = \frac{1}{N} (\hat{\mathbf{y}} - \mathbf{y})^t \odot (\hat{\mathbf{y}} \odot (\mathbf{1} - \hat{\mathbf{y}}))^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{X}^{(4)}} = \frac{1}{N} (\hat{\mathbf{y}} - \mathbf{y})^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial b^{(3)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t} \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \frac{\partial \mathbf{X}^{(3)}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \text{ReLU}'(\mathbf{X}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \mathbf{b}^{(1)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

Backward: cálculos

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \frac{1}{N}(\hat{\mathbf{y}} - \mathbf{y})^t = (0.1250, -0.0673, -0.0673, 0.1828)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t = \begin{pmatrix} 0.2310 \\ 0.1828 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t = 0.1732$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t} \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \begin{pmatrix} 0.1250 & -0.0673 & -0.0673 & 0.1828 \\ -0.1250 & 0.0673 & 0.0673 & -0.1828 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \text{ReLU}'(\mathbf{X}^{(2)t}) = \begin{pmatrix} 0 & -0.0673 & -0.0673 & 0.1828 \\ 0 & 0 & 0 & -0.1828 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ -0.0673 & 0 \\ -0.0673 & 0 \\ 0.1828 & -0.1828 \end{pmatrix} = \begin{pmatrix} 0.1155 & -0.1828 \\ 0.1155 & -0.1828 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t = (1, 1, 1, 1) \begin{pmatrix} 0 & 0 \\ -0.0673 & 0 \\ -0.0673 & 0 \\ 0.1828 & -0.1828 \end{pmatrix} = (0.0482, -0.1828)$$

Actualización de parámetros: capa a capa, respetando el formato (transpuesto) del modelo

$$\mathbf{W}^{(3)t} = \mathbf{W}^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 0.1 \begin{pmatrix} 0.2310 \\ 0.1828 \end{pmatrix} = \begin{pmatrix} 0.9769 \\ -1.0183 \end{pmatrix}$$

$$b^{(3)t} = b^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial b^{(3)}} = 0 - 0.1 \cdot 0.1732 = -0.0173$$

$$\mathbf{W}^{(1)t} = \mathbf{W}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - 0.1 \begin{pmatrix} 0.1155 & -0.1828 \\ 0.1155 & -0.1828 \end{pmatrix} = \begin{pmatrix} 0.9885 & 1.0183 \\ 0.9885 & 1.0183 \end{pmatrix}$$

$$\mathbf{b}^{(1)t} = \mathbf{b}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}} = (0, -1) - 0.1 (0.0482, -0.1828) = (-0.0048, -0.9817)$$

Solución a máquina:

```
In [ ]: import numpy as np; np.set_printoptions(precision=4); import tensorflow as tf; import keras
X1 = keras.Input(shape=(2,))
W1 = tf.constant_initializer([[1, 1], [1, 1]]); b1 = tf.constant_initializer([[0], [-1]])
X3 = keras.layers.Dense(2, activation='relu', kernel_initializer=W1, bias_initializer=b1)(X1)
W3 = tf.constant_initializer([[1, -1]]); b3 = tf.constant_initializer([0])
hy = keras.layers.Dense(1, activation='sigmoid', kernel_initializer=W3, bias_initializer=b3)(X3)
M = keras.Model(inputs=X1, outputs=hy);

In [ ]: X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]]); y = np.array([[0], [1], [1], [0]])
M_X3 = keras.Model(inputs=X1, outputs=M.layers[1].output); print('X3 =', M_X3(X))
print('hy =', M(X))
M.compile(loss='crossentropy', optimizer=keras.optimizers.SGD(learning_rate=0.1))
M.fit(X, y, epochs=1, verbose=1); print(M.layers[1].get_weights(), "\n", M.layers[2].get_weights())

X3 = tf.Tensor(
[[0. 0.]
 [1. 0.]
 [1. 0.]
 [2. 1.]], shape=(4, 2), dtype=float32)
hy = tf.Tensor(
[[0.5    ]
 [0.7311]
 [0.7311]
 [0.7311]], shape=(4, 1), dtype=float32)
1/1 0s 326ms/step - loss: 0.6582
[array([[0.9884, 1.0183],
        [0.9884, 1.0183]], dtype=float32), array([-0.0048, -0.9817], dtype=float32)]
[array([[ 0.9769],
        [-1.0183]], dtype=float32), array([-0.0173], dtype=float32)]
```

7 Aprendizaje profundo

Aprendizaje profundo: área del ML especializada en el aprendizaje de redes con (muchas) capas ocultas

Revolución del aprendizaje profundo: el aprendizaje profundo alcanza gran popularidad a partir de los 2010, principalmente por la mejora de resultados en reconocimiento automático del habla (2011) y clasificación de imágenes (2012)

LeNet (1998): red **convolucional (CNN)** pionera, introducida en 1998 por Yann LeCun para MNIST

AlexNet (2012): CNN de Alex Krizhevsky para ImageNet que atrajo mucho interés por la gran mejora de resultados

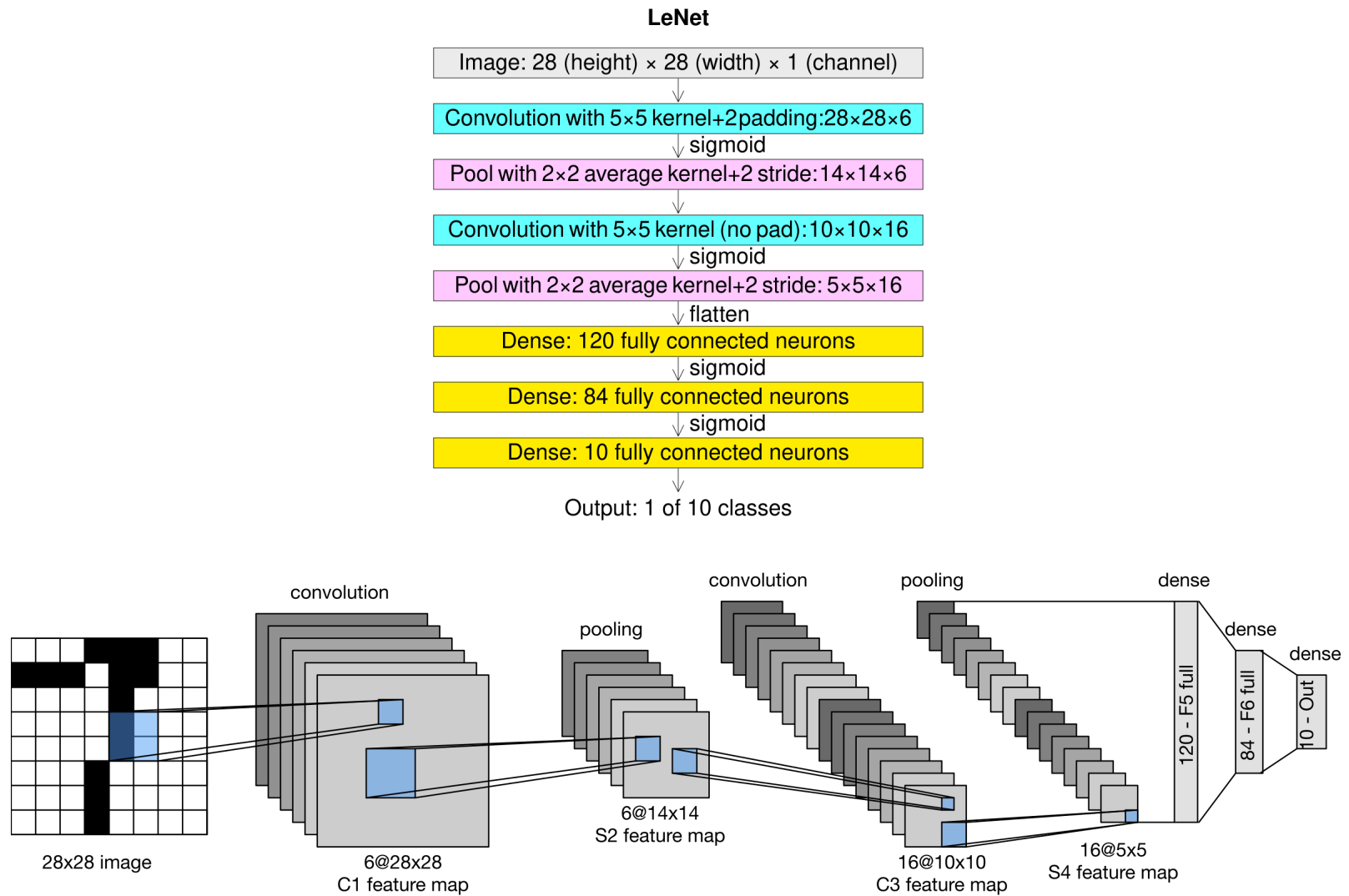
Inception (2014): CNN de Google que introduce la meta-arquitectura típica de las CNNs modernas

ResNet (2015): CNN de Microsoft que introduce **conexiones residuales** para entrenar modelos muy profundos

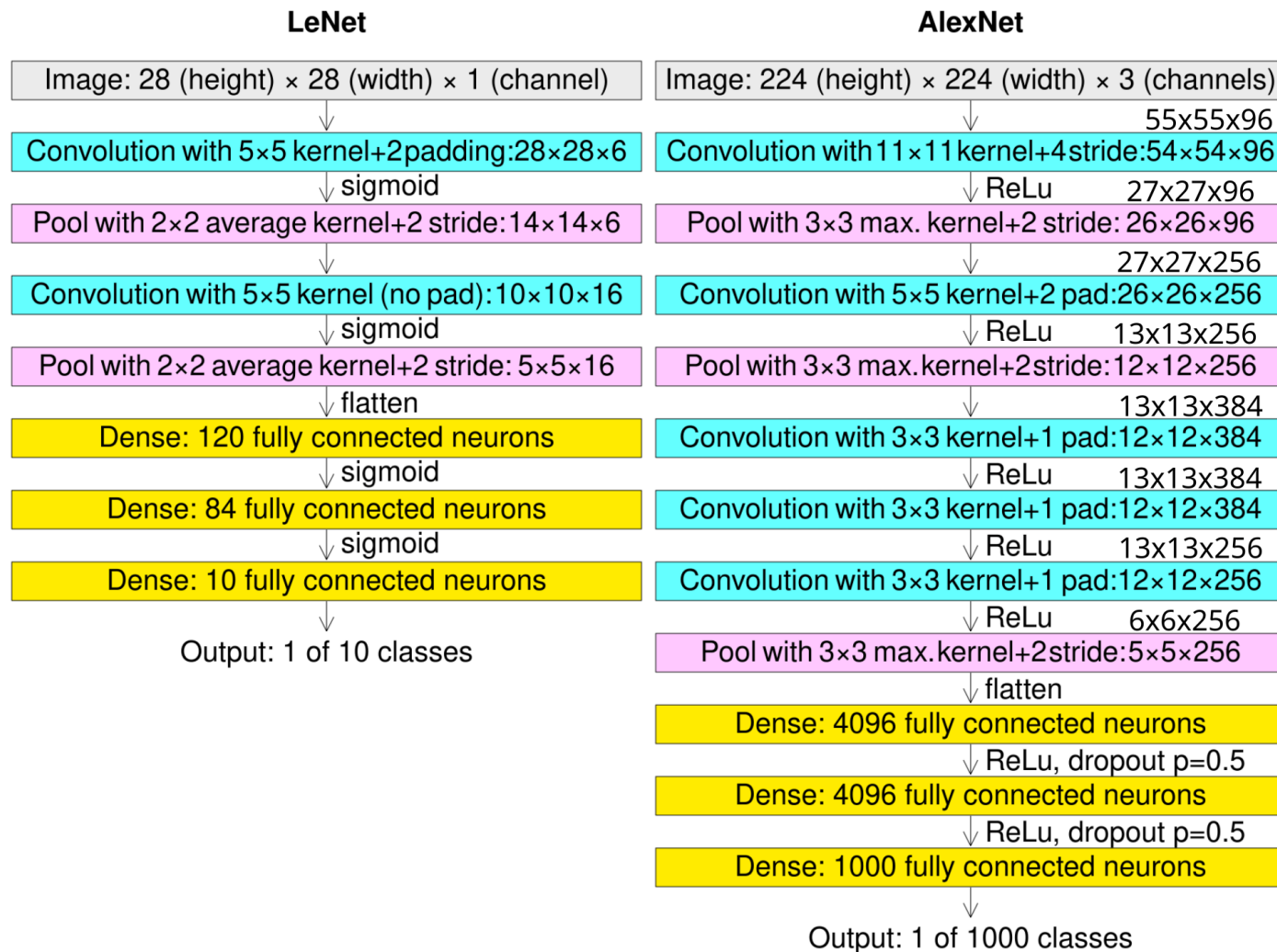
Transformer (2017): arquitectura de Google para traducción automática basada en el **mecanismo de atención**

Vision Transformer (2020): adaptación del Transformer para visión que rivaliza con las CNNs

LeNet (1998): red **convolucional (CNN)** pionera, introducida en 1998 por Yann LeCun para MNIST

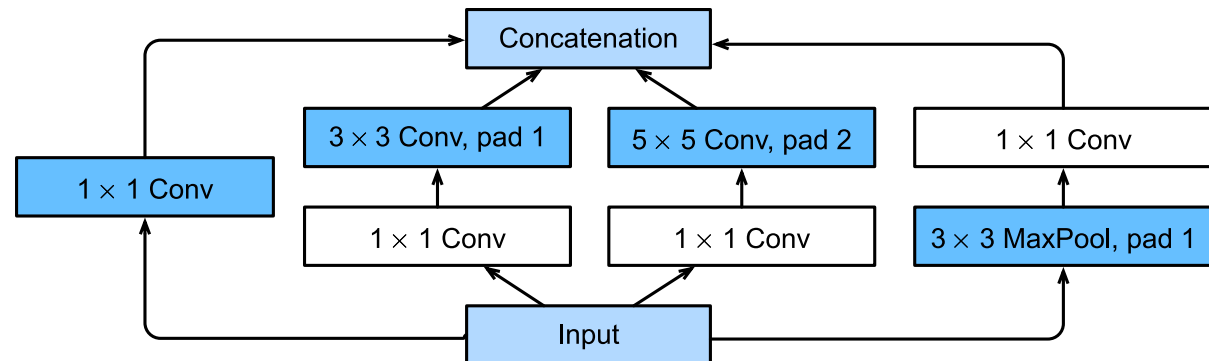
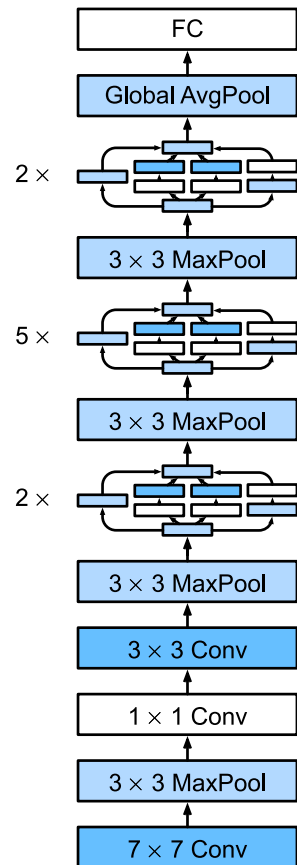


AlexNet (2012): CNN de Alex Krizhevsky para ImageNet que atrajo mucho interés por la gran mejora de resultados

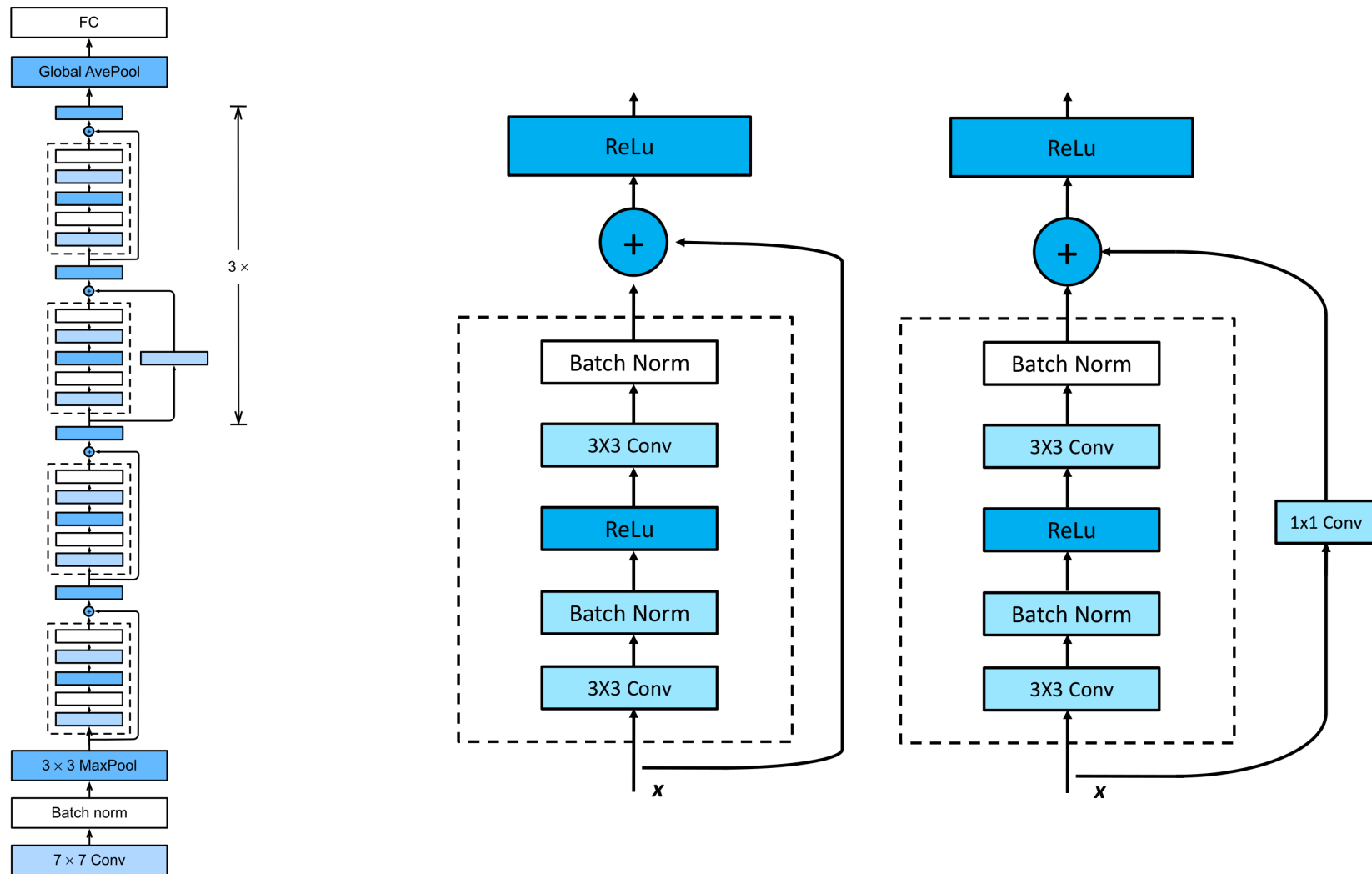


Inception (2014): CNN de Google que introduce la meta-arquitectura típica de las CNNs modernas

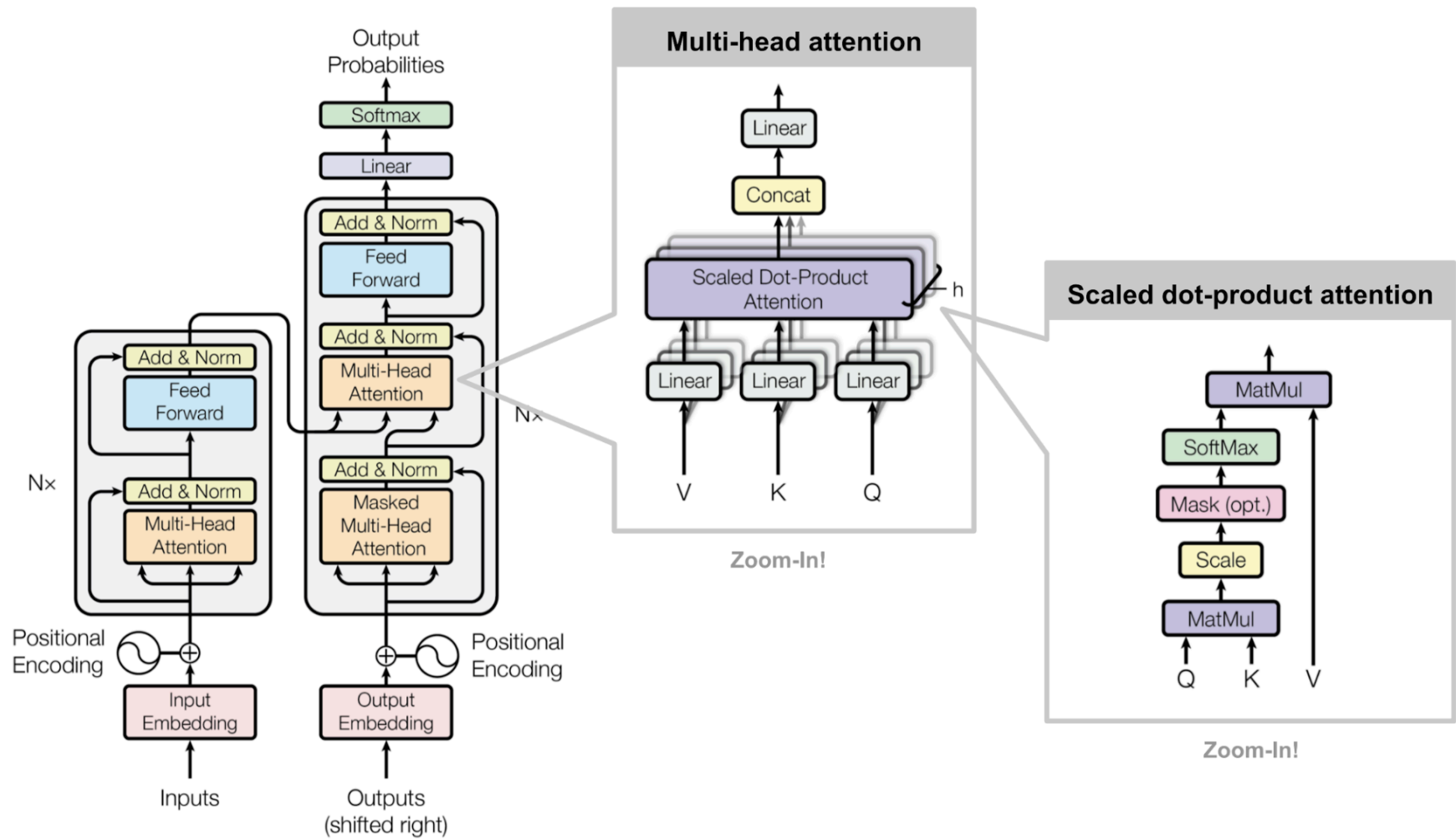
- **Stem (raíz):** dos o tres capas convolucionales que extraen características de bajo nivel
- **Body (cuerpo):** subred de **bloques convolucionales** repetidos
- **Head (cabeza):** transforma la salida del cuerpo según la tarea a abordar (clasificación, segmentación, etc.)



ResNet (2015): CNN de Microsoft que introduce **conexiones residuales** para entrenar modelos muy profundos

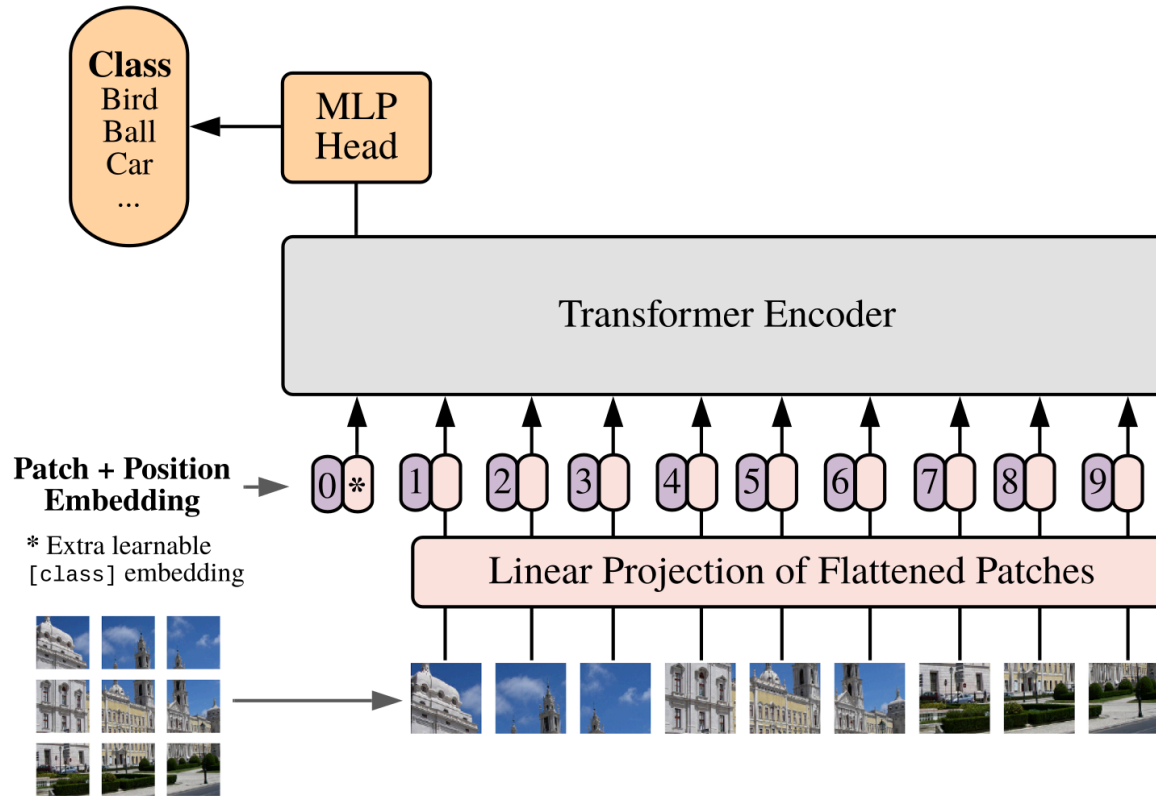


Transformer (2017): arquitectura de Google para traducción automática basada en el **mecanismo de atención**



Vision Transformer (2020): adaptación del Transformer para visión que rivaliza con las CNNs

Vision Transformer (ViT)



Transformer Encoder

